

BracketWise Engine - Calculation & Compliance Map

0. Core Parameters

Federal (`FED_BRACKETS` , `BPA_PARAMS` , `CEA_PARAMS` , `MEDICAL_FLOOR`)

Parameter	2025	2026	CRA Reference
NR credit rate (<code>nrcRate</code>)	14.5%	14%	First bracket rate — read dynamically
BPA max (<code>fedMax</code>)	\$16,129	\$16,452	Line 30000
BPA min (<code>fedMin</code>)	\$14,538	\$14,829	Line 30000 (high earners)
BPA phase-out starts	\$177,882	\$181,441	
BPA phase-out ends	\$253,414	\$258,483	
Canada Employment Amount	\$1,469	\$1,498	Line 31260
Medical expense floor	\$2,834	\$2,890	Line 33099

CPP (`CPP_PARAMS`)

Symbol	Meaning	2025	2026
<code>ympe</code>	Tier 1 ceiling	\$71,300	\$74,600
<code>yampe</code>	Tier 2 ceiling	\$81,200	\$85,000
<code>basicEx</code>	Basic annual exemption	\$3,500	\$3,500
<code>emCPP1</code>	Employee/employer CPP1 rate	5.95%	5.95%
<code>seCPP1</code>	SE CPP1 rate (both halves)	11.90%	11.90%
<code>emMaxCPP1</code>	Max employee CPP1	\$4,034.10	\$4,230.45
<code>seMaxCPP1</code>	Max SE CPP1 (both halves)	\$8,068.20	\$8,460.90
<code>emCPP2</code>	Employee/employer CPP2 rate	4.00%	4.00%
<code>seCPP2</code>	SE CPP2 rate (both halves)	8.00%	8.00%
<code>emMaxCPP2</code>	Max employee CPP2 = $(\text{yampe} - \text{ympe}) \times \text{emCPP2}$	\$396.00	\$416.00
<code>seMaxCPP2</code>	Max SE CPP2 (both halves)	\$792.00	\$832.00
<code>CPP1_BASE_RATE</code>	Statutory base rate (module constant)	4.95%	4.95%

EI (`EI_PARAMS`) — T4 employee only; SE EI excluded

	2025	2026
Rate	1.64%	1.63%
Max insurable earnings	\$65,700	\$68,900
Max annual premium	\$1,077.48	\$1,123.07

1. Income Aggregation

Variable	Formula	Notes
<code>cgInclusion</code>	$\text{capitalGains} \leq \$250K \rightarrow \times 0.5$; above → $\$125,000 + \text{excess} \times 2/3$	Tiered inclusion — not a flat 50%

Variable	Formula	Notes
<code>taxableGrossIncome</code>	<code>t4Income + seNetIncome + otherTaxableIncome + cgInclusion</code>	Capital gains counted at inclusion rate; feeds <code>netIncome</code>
<code>grossIncome</code>	<code>t4Income + seNetIncome + otherTaxableIncome + capitalGains</code>	Full economic value; used for display and <code>avgTaxRate</code> denominator only

2. T4 — CPP Calculations

Tier 1

Variable	Formula	CRA Line
<code>t4Pensionable1</code>	<code>max(0, min(t4Income, ympe) - basicEx)</code>	Schedule 8
<code>t4Cpp1</code>	<code>min(t4Pensionable1 × emCPP1, emMaxCPP1)</code>	Schedule 8
<code>t4CppNrtcBase</code>	<code>t4Cpp1 × (CPP1_BASE_RATE / emCPP1)</code> — the 4.95% base portion	30800
<code>t4EnhancedDeduction</code>	<code>t4Cpp1 - t4CppNrtcBase</code> — the enhanced 1% portion	22215

Tier 2

Variable	Formula	CRA Line
<code>t4Cpp2Pensionable2</code>	<code>max(0, min(t4Income, yampe) - ympe)</code>	Schedule 8
<code>t4Cpp2</code>	<code>min(t4Cpp2Pensionable2 × emCPP2, emMaxCPP2)</code>	Schedule 8

Combined T4 income deduction fed into `netIncome` :

Variable	Formula	CRA Line
<code>t4CppLine22215</code>	<code>t4EnhancedDeduction + t4Cpp2</code>	22215

3. SE — CPP Calculations

T4 income consumes the CPP room first; SE only contributes to what remains.

Tier 1 — Room Allocation

Variable	Formula
<code>totalCombinedPensionable1</code>	<code>min(t4Income + seNetIncome, ympe)</code>
<code>seAllowedEarnings1</code>	<code>max(0, totalCombinedPensionable1 - max(0, t4Income))</code>
<code>t4UnderExemptionBonus</code>	<code>max(0, basicEx - min(t4Income, basicEx))</code> — unused exemption carried from T4
<code>seContributoryEarnings1</code>	<code>max(0, seAllowedEarnings1 - t4UnderExemptionBonus)</code>
<code>remainingSeMax1</code>	<code>seMaxCPP1 - T4's consumed equivalent</code> — prevents exceeding annual cap
<code>seCpp1</code>	<code>min(seContributoryEarnings1 × seCPP1, remainingSeMax1)</code>

Tier 1 — Line Allocations

Variable	Formula	CRA Line
<code>seEmployerDeduction</code>	<code>seCpp1 × 0.5</code>	22200
<code>seNrtcBase</code>	<code>seCpp1 × (CPP1_BASE_RATE / seCPP1)</code> — the 4.95% base portion	31000

Variable	Formula	CRA Line
(part of <code>seEnhancedDeduction</code>)	<code>seCpp1 × (enhR1 / seCPP1)</code> — the enhanced 1% portion	22215

Tier 2

Variable	Formula	CRA Line
<code>totalCombinedPensionable2</code>	<code>min(t4Income + seNetIncome, yampe)</code>	Schedule 8
<code>t4YampeEarningsConsumed</code>	<code>max(0, min(t4Income, yampe) - ympe)</code>	
<code>seAllowedEarnings2</code>	Tier 2 band total minus T4's share	
<code>seCpp2</code>	<code>min(seAllowedEarnings2 × seCPP2, remainingSeMax2)</code>	Schedule 8

Combined SE income deduction fed into `netIncome` :

Variable	Formula	CRA Lines
<code>seEnhancedDeduction</code>	<code>seCpp1 × (enhR1 / seCPP1) + seCpp2</code>	22215
<code>seCppDeductions</code>	<code>seEmployerDeduction + seEnhancedDeduction</code>	22200 + 22215

4. EI

Variable	Formula	CRA Line
<code>ei</code>	<code>min(min(t4Income, maxInsurable) × rate, maxPremium)</code>	31200

5. Net Income

```
netIncome = max(0,
taxableGrossIncome
- seCppDeductions      Lines 22200 + 22215
- t4CppLine22215       Line 22215
- rrspAlreadyContributed Line 20800
- rrspContrib           Line 20800
- fhscAlreadyThisYear   Line 20805
- fhscContrib           Line 20805
- childcare             Line 21400
)
```

`netIncome` is CRA Lines 23600 and 26000 — they're the same in this app because no deductions exist between them (no LCGE, loss carryforwards, etc.). It is the amount fed into all bracket calculations.

6. Federal Tax

Variable	Formula	CRA Line
<code>rawFed</code>	<code>bracketedTax(netIncome, FED_BRACKETS[year])</code>	
<code>bpa</code>	Scales linearly <code>fedMax → fedMin</code> between the phase-out thresholds	30000
<code>bpaCred</code>	<code>bpa × nrcRate</code>	30000
<code>cppNrtc</code>	<code>(t4CppNrtcBase + seNrtcBase) × nrcRate</code>	30800 / 31000
<code>eiCred</code>	<code>ei × nrcRate</code>	31200
<code>ceaCred</code>	<code>min(t4Income, CEA_PARAMS[year]) × nrcRate</code>	31260
<code>medFloor</code>	<code>min(netIncome × 3%, MEDICAL_FLOOR[year])</code>	33099
<code>medCred</code>	<code>max(0, medExp - medFloor) × nrcRate</code>	33099

Variable	Formula	CRA Line
fedTax	$\max(0, \text{rawFed} - \text{bpaCred} - \text{cppNrtc} - \text{eiCred} - \text{ceaCred} - \text{medCred})$	42000

7. Provincial Tax

Variable	Formula	Notes
provCppNrtcBase	$\text{t4CppNrtcBase} + \text{seNrtcBase}$	Shared base for provincial CPP credit
rawProv	$\text{bracketedTax}(\text{netIncome}, \text{PROV_BRACKETS}[\text{year}][\text{province}])$	
provCppEiCred	$(\text{provCppNrtcBase} + \text{ei}) \times \text{lowestProvRate}$	BC: 5.6%; AB: 8%; SK: 10.5% (2026)
BC LITR	Full credit below \$25,570; 3.56% clawback to \$44,952; \$0 above (2026)	BC_CLAWBACK
provTax	$\max(0, \text{rawProv} - \text{provBpaCred} - \text{provCppEiCred} - \text{litR})$	
provMedFloor	$\min(\text{netIncome} \times 3\%, \text{PROV_MEDICAL_FLOOR}[\text{year}][\text{province}])$	Provincial floor follows the same CRA lesser-of rule as federal
provMedCred	$\max(0, \text{medExp} - \text{provMedFloor}) \times \text{lowestProvRate}$	Applied after calcProv
adjProvTax	$\max(0, \text{provTax} - \text{provMedCred})$	Final provincial tax

8. Optimization Cascade

The engine uses a **5-pass convergence loop** rather than a single forward pass, because the optimization is self-referential: contributing to RRSP reduces `netIncome`, which reduces `totalTax`, which reduces `taxOwing`, which increases `investablePool`, which enables a larger RRSP contribution — requiring the loop to re-solve until stable.

Each pass:

1. Run `singlePass` with current `rrspContrib` and `fhsaContrib` to get updated tax results
2. Recompute `taxOwing` and `seCpp` based on that pass's numbers — never on pre-optimization figures
3. Derive `investablePool = max(0, availableCash - taxOwing - seCpp)` — cash that is genuinely safe to invest
4. Set `fhsaContrib = min(investablePool, fhsaRoom)` — Priority 1
5. Compute `rrspHeadroom` — the exact distance between the current `netIncome` and the province's tax-free floor, adjusted for any FHSA shift in this pass
6. Set `rrspContrib = min(remaining, rrspRoom, rrspHeadroom)` — Priority 2, capped at floor

After 5 passes the contributions are stable. A final sixth `singlePass` is then run with the converged values to produce the reported result.

Variable	Formula	Notes
taxOwing	$\max(0, \text{totalTax} - \text{t4TotalTax})$ per pass	Recalculated each pass — never frozen at pre-contribution value
investablePool	$\max(0, \text{availableCash} - \text{taxOwing} - \text{seCpp})$	Strictly safe cash — cannot over-commit to contributions
fhsaContrib	$\min(\text{investablePool}, \text{fhsaRoom})$	Priority 1
rrspHeadroom	$\text{lastPass.netIncome} + \text{rrspContrib}_{i-1} + \text{fhsaContrib}_{i-1} - \text{fhsaContrib}_i - \text{floor}$	Distance to the provincial optimization floor

Variable	Formula	Notes
<code>rrspContrib</code>	<code>min(remaining, rrspRoom, rrspHeadroom)</code>	Priority 2 — hard stop at floor
<code>tfsa</code>	<code>min(max(0, investablePool - fhsaContrib - rrspContrib), tfsaAnnualLimit)</code>	Capped at annual limit
<code>isBpaWasted</code>	<code>result.netIncome < floor</code>	Flag: user was already below the floor; no RRSP headroom existed

FHSA room = `min(fhsaAnnual, fhsaLifetime - fhsaCumulativeContributed - fhsaAlreadyThisYear)`

RRSP headroom = room to reduce `netIncome` to the tax-free floor (`TAX_FREE_FLOOR`) for the province, accounting for employer match impact.

▼ Case study

Inputs

Province / Year	BC · 2026
T4 Income	\$56,000
SE Net Income	\$35,000
Capital Gains	\$12,000
Other Income	\$2,000
Available Cash	\$2,500/mo → \$30,000/yr
FHSA room for year	\$16,000
FHSA contributed already	\$2,500
RRSP room (from NOA)	\$27,000
RRSP already contributed	\$0
RRSP match %	0%

Step 1 — Income Aggregation

	Formula	Result
Capital gains inclusion	$12,000 \times 50\% (\leq \$250K \text{ threshold})$	\$6,000
<code>taxableGrossIncome</code>	$56,000 + 35,000 + 2,000 + 6,000$	\$99,000
<code>grossIncome</code>	$56,000 + 35,000 + 2,000 + 12,000$	\$105,000

Step 2 — CPP

T4 — Tier 1

	Formula	Result
<code>t4Pensionable1</code>	$\min(56,000, 74,600) - 3,500$	\$52,500
<code>t4Cpp1</code>	$\min(52,500 \times 5.95\%, 4,230.45)$	\$3,123.75

	Formula	Result
t4CppNrtcBase (→ Line 30800)	$3,123.75 \times (4.95 / 5.95)$	\$2,598.75
t4EnhancedDeduction (→ Line 22215)	$3,123.75 - 2,598.75$	\$525.00

T4 — Tier 2

	Formula	Result
t4Cpp2Pensionable2	$\min(56,000, 85,000) - 74,600 = \text{negative}$	\$0
t4Cpp2	—	\$0
t4CppLine22215 total	$525.00 + 0$	\$525.00

T4 income is below the YMPE (\$74,600) so no Tier 2 T4 CPP.

SE — Tier 1

	Formula	Result
totalCombinedPensionable1	$\min(56,000 + 35,000, 74,600)$	\$74,600
seAllowedEarnings1	$74,600 - 56,000$	\$18,600
t4UnderExemptionBonus	$3,500 - \min(56,000, 3,500) = 3,500 - 3,500$	\$0
seContributoryEarnings1	$18,600 - 0$	\$18,600
remainingSeMax1	$8,460.90 - (52,500 \times 11.9\%) = 8,460.90 - 6,247.50$	\$2,213.40
seCpp1	$\min(18,600 \times 11.9\%, 2,213.40)$	\$2,213.40

SE — Tier 2

	Formula	Result
totalCombinedPensionable2	$\min(56,000 + 35,000, 85,000)$	\$85,000
totalYampeEarningsAllocated	$85,000 - 74,600$	\$10,400
t4YampeEarningsConsumed	$\min(56,000, 85,000) - 74,600 = \text{negative}$	\$0
seAllowedEarnings2	$10,400 - 0$	\$10,400
seCpp2	$\min(10,400 \times 8\%, 832.00)$	\$832.00

SE income (\$35,000) carries all of the YAMPE band because T4 never reached YMPE.

SE — Line Allocations

CRA Line	Variable	Formula	Result
22200	seEmployerDeduction	$2,213.40 \times 50\%$	\$1,106.70
22215	seEnhancedDeduction	$2,213.40 \times (1\%/11.9\%) + 832$	\$1,018.00
31000	seNrtcBase	$2,213.40 \times (4.95\%/11.9\%)$	\$920.70
—	seCppDeductions	$1,106.70 + 1,018.00$	\$2,124.70

CPP Summary

Stream	Amount
T4 CPP1 (employee)	\$3,123.75
T4 CPP2 (employee)	\$0.00
SE CPP1 (both halves)	\$2,213.40
SE CPP2 (both halves)	\$832.00

Stream	Amount
Total CPP	\$6,169.15

Step 3 — EI

$ei = \min(56,000 \times 1.63\%, 1,123.07) = \min(\$912.80, \$1,123.07) = \mathbf{\$912.80}$

T4 income is below the \$68,900 insurable ceiling, so EI is not maxed.

Step 4 — T4 Withholding Estimate

`estimateT4Withholding` runs a T4-only `singlePass` (no SE, no FHSA, no RRSP):

T4-only net income	56,000 – 525 (Line 22215) = \$55,475
Raw federal	55,475 × 14% = \$7,766.50
Credits (BPA + CPP + EI + CEA)	–\$3,004.62
Federal tax	\$4,761.88
Raw BC provincial	50,363 × 5.6% + 5,112 × 7.7% = \$3,213.95
BC credits (BPA + CPP/EI)	–\$936.75
BC LITR	\$0 (net income > \$44,952)
BC provincial tax	\$2,277.20
t4TotalTax	\$7,039.08

Step 5 — Available Room

	Formula	Result
<code>fhsaRoom</code>	$\min(16,000 - 2,500, 40,000 - 0)$	\$13,500
<code>rrspRoom</code>	$27,000 - 0 - 0$	\$27,000
BC optimization floor	<code>TAX_FREE_FLOOR.BC</code> (fixed)	\$21,000

Step 6 — 5-Pass Convergence Loop

Available cash = **\$30,000/yr.** `seCpp` (cash owed in April) = \$3,045.40 (unchanged across passes).

Pass	netIncome	totalTax	taxOwing	investablePool	fhsaContrib	rrspContrib
0 (seed)	\$93,850	\$17,482	\$10,443	\$16,511	\$13,500	\$3,011
1	\$77,339	\$12,826	\$5,787	\$21,168	\$13,500	\$7,668
2	\$72,683	\$11,512	\$4,473	\$22,482	\$13,500	\$8,982
3	\$71,369	\$11,142	\$4,103	\$22,851	\$13,500	\$9,351
4	\$70,999	\$11,038	\$3,999	\$22,955	\$13,500	\$9,455

`rrspHeadroom` in all passes = \$59,350 (far above the cash limit; never the binding constraint). The RRSP is **cash-limited**, not room-limited.

Step 7 — Final Pass

With `fhsaContrib = $13,500`, `rrspContrib = $9,455`:

Net Income

Deduction	Amount
seCppDeductions (Lines 22200 + 22215)	-\$2,124.70
t4CppLine22215 (Line 22215)	-\$525.00
FHSA already this year (Line 20805)	-\$2,500.00
FHSA optimizer (Line 20805)	-\$13,500.00
RRSP optimizer (Line 20800)	-\$9,455.44
Net Income (Line 23600)	\$70,895

Federal Tax

rawFed	$58,524 \times 14\% + 12,371 \times 20.5\% = \mathbf{\$10,729}$
BPA credit ($16,452 \times 14\%$)	-\$2,303
CPP NR credit ($3,519.45 \times 14\%$)	-\$493
EI credit ($912.80 \times 14\%$)	-\$128
CEA credit ($1,498 \times 14\%$)	-\$210
Federal Tax	\$7,596

BC Provincial Tax

rawProv	$50,363 \times 5.6\% + 20,532 \times 7.7\% = \mathbf{\$4,401}$
BC BPA credit ($13,216 \times 5.6\%$)	-\$740
BC CPP+EI credit ($4,432 \times 5.6\%$)	-\$248
BC LITR ($\$70,895 > \$44,952$)	\$0
BC Provincial Tax	\$3,413

Results (what the user sees)

Income

Gross Income	\$105,000
Taxable Income (net, after deductions)	\$70,895
After-Tax Income	\$86,909
Average Tax Rate	17.2%

Tax Breakdown

Federal Tax	\$7,596
BC Provincial Tax	\$3,413
Total Income Tax	\$11,009
Estimated T4 Withholding	\$7,039
Tax Owing in April	\$3,970
Tax Saving (vs. no optimization)	\$6,473

CPP & EI

T4 CPP1	\$3,124
T4 CPP2	\$0
SE CPP1	\$2,213
SE CPP2	\$832
Total CPP	\$6,169
EI	\$913

April Cash Obligations

Tax owing	\$3,970
SE CPP (not withheld at source)	\$3,045
Total Liabilities	\$7,015

Optimization Allocations (of \$30,000 available)

FHSA (optimizer)	\$13,500
FHSA (already contributed)	\$2,500
Total FHSA this year	\$16,000 (full room used)
RRSP	\$9,455 (cash-limited; \$17,545 room unused)
TFSA	\$29
April liabilities	\$7,015
Total	\$30,000

Marginal Rates

Before optimization	28.2%
After optimization	28.2%

Both before and after sit in the same bracket pair (20.5% federal + 7.7% BC). The optimization reduced income within the same tier.

Chart

Before cursor (taxable income)	\$93,850
After cursor (taxable income)	\$70,895
Income shielded zone width	\$22,955

Alerts

GST registration threshold	⚠ SE income \$35,000 exceeds \$30,000 — GST registration required
Cash deficit	No

BPA wasted (income below \$21K floor)	No
BC LITR clawback zone	No (net income \$70,895 > \$44,952 ceiling)